

I wish the camera captured all the colours as beautifully as they look in real life.

I know what you mean, but we can fix the colours and the light in a picture using an image editing software.



The colours in an image depend on various factors at the time it was captured. Fog, smoke, sunlight, clouds, etc. all affect an image. At times, an image does not look like what we see in real life. In this case, we can use colour correction to edit the colours in an image.

Recall:

Every image is made of lots of tiny boxes called pixels. Each pixel contains a colour. The colour in a pixel can be stored by using a combination of red, green, and blue (RGB).

What is Colour Correction?

Colour Correction is a technique used to make colours in an image look better. Issues with colours are fixed to make the image look as natural as possible. Colour correction adjusts the brightness, exposure, contrast, white balance, etc. of an image.

Activity 1

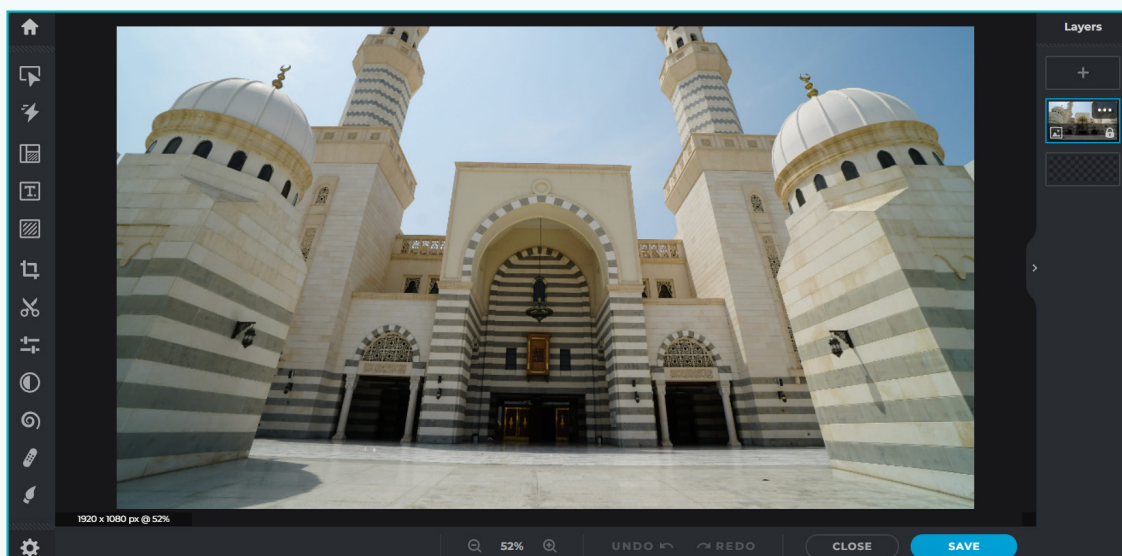
Colour correct an image in the Pixlr image editor.



Original

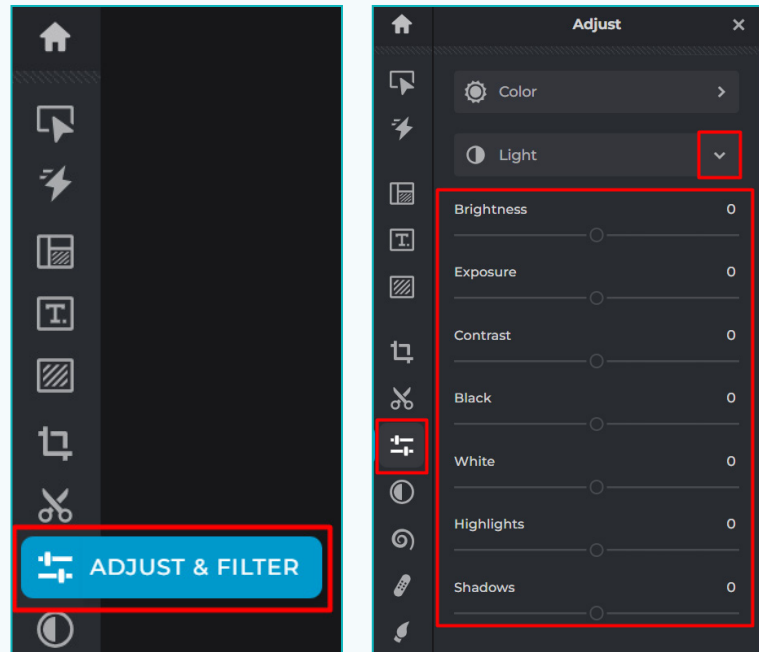
Colour correction

1 Open the image in **Pixlr**.



Activity 1

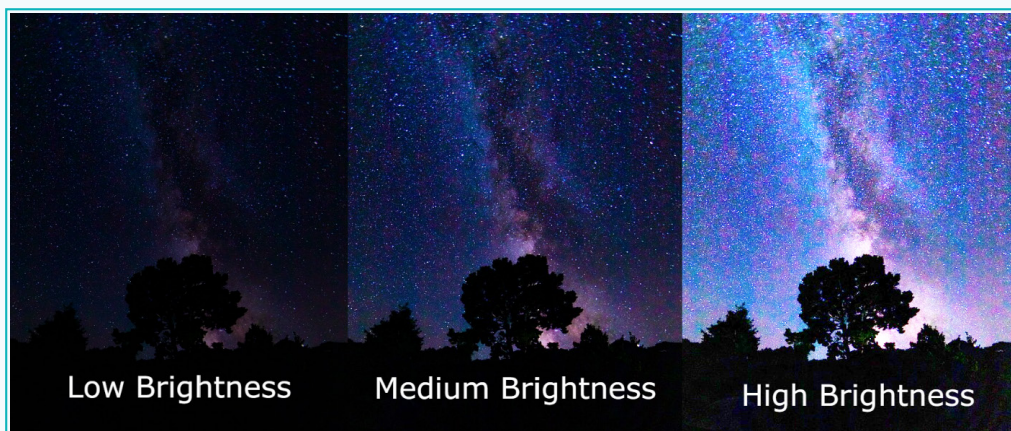
- 2 Click on Adjust and Filter. Expand the 'Light' category.



The Light category has 3 main parameters:

- **Brightness**

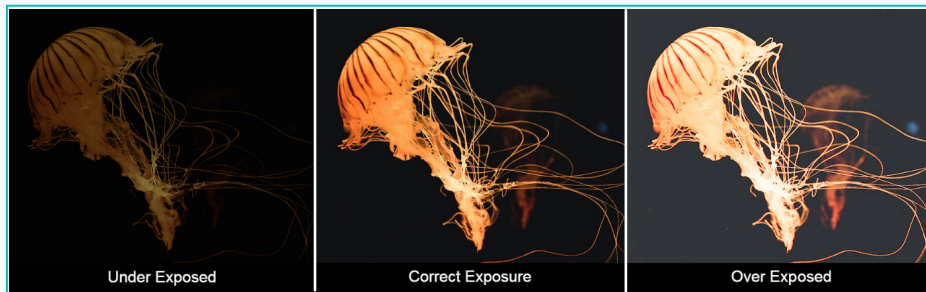
When the brightness of an image is increased, it increases the light in an image to make it appear brighter. When the brightness of an image is decreased, it decreases the light in an image to make it appear darker.



Activity 1

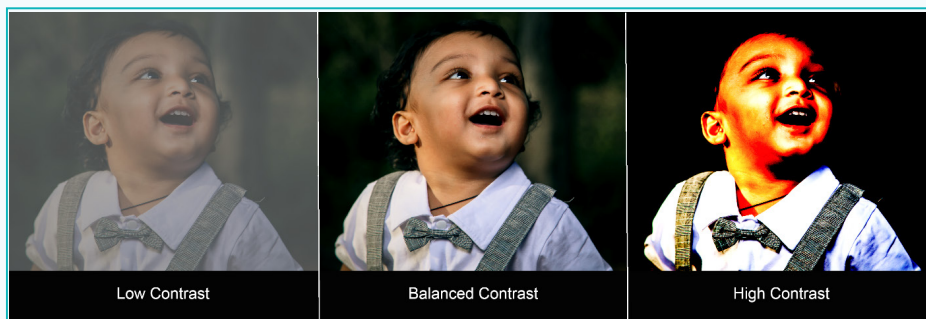
- **Exposure**

The exposure determines the amount of light in an image. When the exposure is changed, it affects the overall glow of an image.

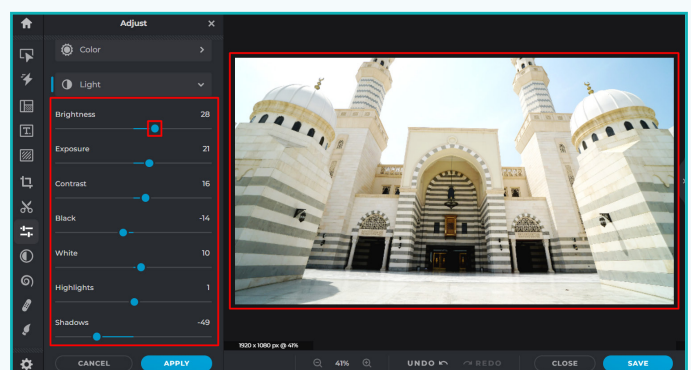


- **Contrast**

Contrast is the difference in colour or light between areas in an image. When the contrast is changed, it affects different areas of the image. The area with more light becomes brighter and the area with less light becomes darker.

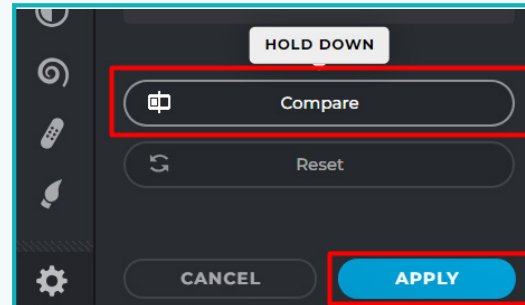


- 3 Use the sliders to tweak the values to make the image appear natural.

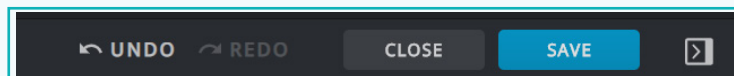


Activity 1

- 4 To observe the changes, click on Compare. Click on 'Apply' to make changes to the image.



- 5 Undo or redo changes with the 'undo' and 'redo' options.



Exercise

Fill in the blanks

- 1 _____ is a technique used to adjust the brightness, exposure, contrast, white balance, etc.
- 2 Changing the _____ value affects the overall light of an image.
- 3 _____ is defined as the difference between various areas in an image.

9. Enhance Images

What is Colour Grading?

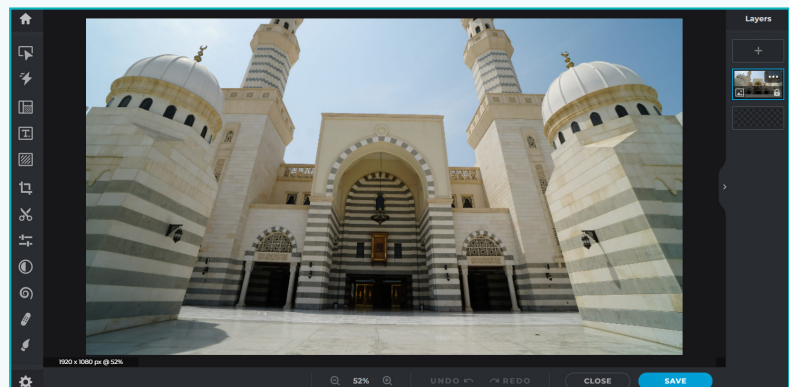
Colour grading is a process that adds mood and feel to an image that makes it feel unnatural. This technique is used to make colours in an image look artificial.

Vibrance, hue, tint, etc. are adjusted in colour grading.

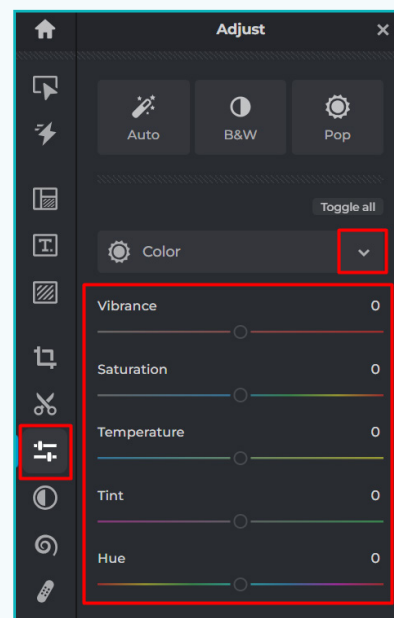
Activity 2

Colour grade an image in the Pixlr image editor.

1 Open the image in **Pixlr**.



2 Click on Adjust and Filter. Expand the 'Color' category.



Activity 2

The following parameters are adjusted to correct colours in an image:

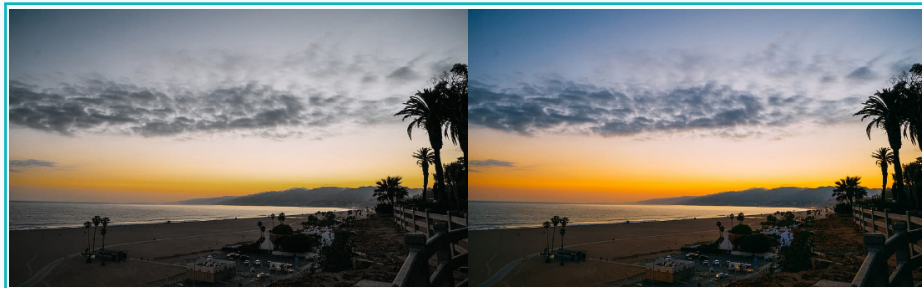
- **Saturation**

Saturation affects the overall intensity of colours in an image.



- **Vibrance**

Vibrance is a smart tool that changes the value of only dull or dark colours in an image and leaves the bright colours alone.



- **Temperature**

Every light source gives off a different colour temperature. Some light sources are warm, and some are cold. When the temperature values are changed, it affects the overall temperature of an image (cold or warm).



Activity 2

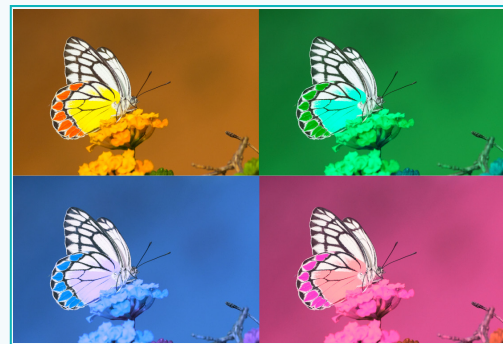
- **Tint**

Tint adds a different colour to an image to modify it. It adds colour tints over an image. The tints range from green to magenta.

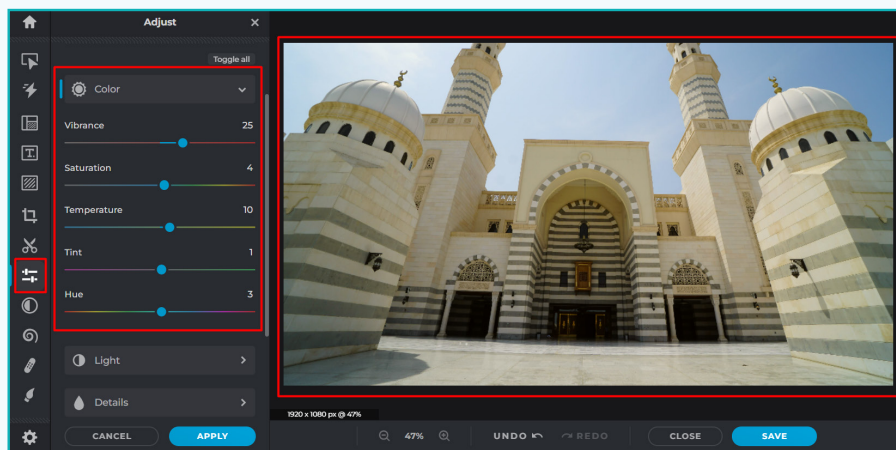


- **Hue**

Hue is one of the main properties of colour in an image. When the hue is changed, the value of each pixel is changed. It replaces existing colours with other colours.



3 Adjust the sliders for these properties to see the changes in an image.

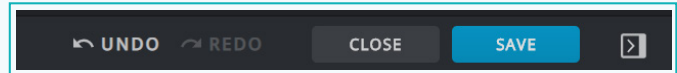


4 Click on 'Apply' once the desired colours are achieved.



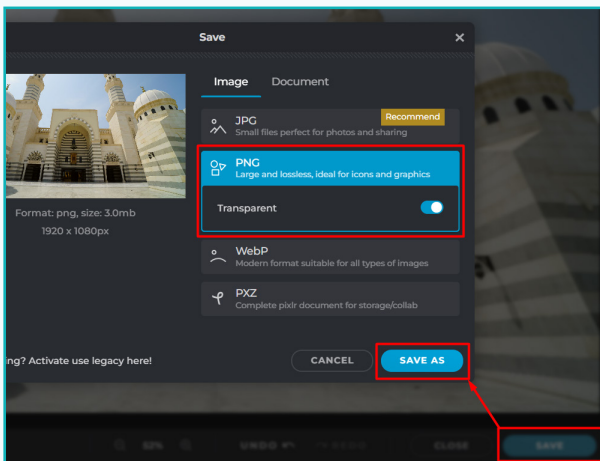
Activity 2

- 5 Undo or redo changes in the images with the undo and redo options.



Note: The values of different parameters of an image depend on the image type. Each image requires different values for each parameter to fix the colour issues.

- 6 Save the image as 'PNG' file type after applying all the changes in the image.



Recall:

There are different types of image files:

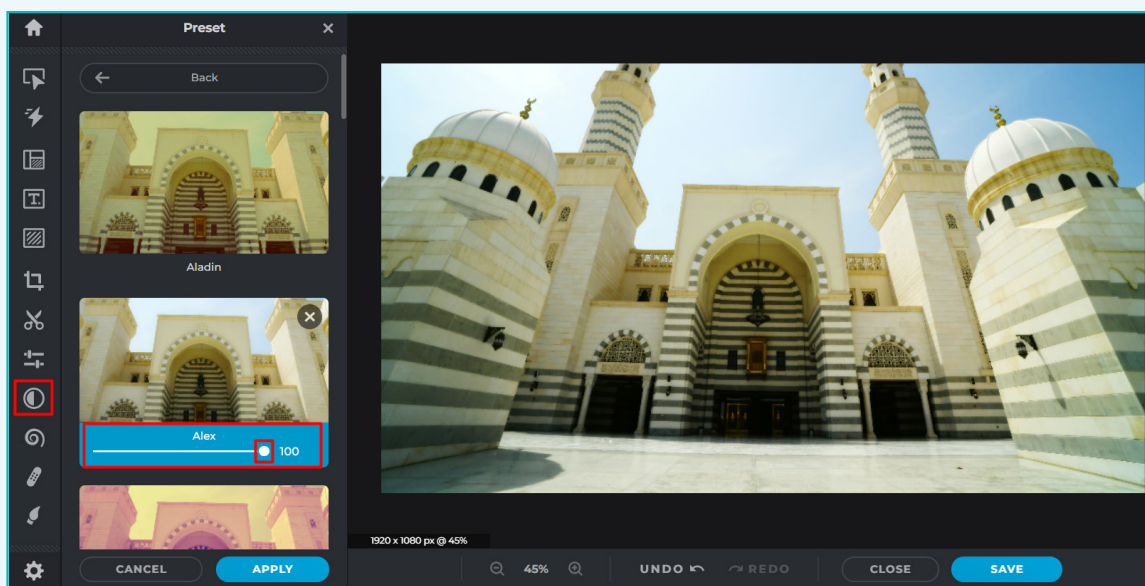
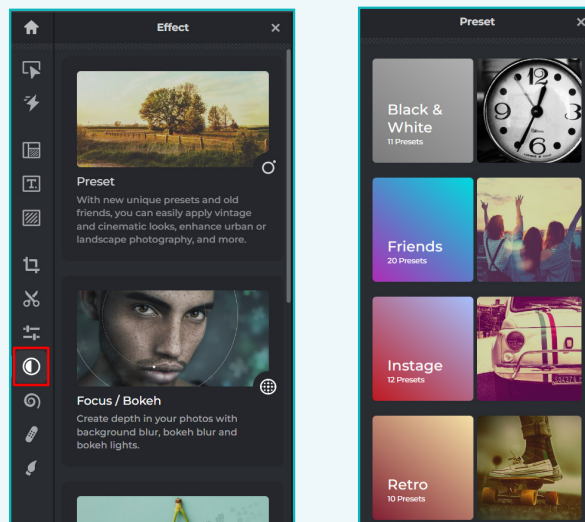
- **JPEG (.jpg, .jpeg):** JPEG stands for Joint Photographic Experts Groups. JPEG is a popular format for digital cameras. This makes it ideal for use on the internet. There is a slight loss in quality when we save an image as .jpg, but it is not very apparent when the image is displayed on screen.
- **PNG (.png):** PNG stands for Portable Network Graphics. PNG files can handle up to 16 million colours. There is no quality loss when we save an image in the .png format.
- **PXD:** Pixlr Editor has its own file format called PXD that preserves image edits to modify later, if required.

Activity 3

Use effects to colour correct an image.

Effects are used to instantly transform pictures. Effects are predefined settings to achieve a certain look and feel in an image.

Effects are split into multiple categories. Each category has multiple predefined colour grading values that can be applied to an image.



The value of the effect can be adjusted using the slider control under each Effect.

Exercise

Fill in the blanks

- 4 _____ affects the overall intensity of colours in the image.
- 5 _____ setting is used to adjust the colour temperature.
- 6 _____ are predefined settings to achieve a certain look and feel for the image.
- 7 _____ is a technical process in which atmosphere and tone are adjusted.
- 8 When the brightness is increased, pixels appear _____. When it is decreased, the pixels appear _____.

Colour enhancement is used to fix colours in a picture. It helps make a bland image look more realistic. Colour grading and colour correction techniques are used to fix the image.



Original



Colour Correction



Colour Grading



Tech Flash

- **Colour Correction** : Colour Correction is a technique that fixes colour issues and makes an image appear as natural as possible.
- **Vibrance** : Vibrance is a smart tool that increases the intensity of the more muted colours and leaves the well-saturated colours alone.
- **Colour Grading** : Colour Grading is a technical process in which atmosphere and tone are adjusted.
- **Saturation** : Saturation describes the depth or intensity of colour in an image.
- **Effects** : Effects are used to instantly transform the image. They have predefined settings.
- **Contrast** : Contrast is defined as the separation between the darkest and brightest areas of an image.